P1. Assignment 1.0. Personal Computer Security Posture Analysis

Determine your computer security posture

Objective

This assignment is intended to determine the security posture of your computer. To do so we will focus in the security protections (Countermeasures) that you have configured in your operating system. Because we will be only covering the more important countermeasures applied, they will be available in whatever your operating system is (Windows, MacOS or Linux). There are many security hardening guides you can use, here you have a few simple ones that can give you ideas about how to improve your computer security posture.

- MS Windows 10 Hardening Guide 1Links to an external site.
- MS Windows 11 Hardening Guide 2Links to an external site.
- MS Windows 10 and 11 hardening guideLinks to an external site.
- MAC Hardening Guide 1Links to an external site.
- MAC Hardening Guide 2 (section 6)Links to an external site.
- Linux Hardening GuideLinks to an external site.

Security configuration aspects to review

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            - **System users**
                - How many users are created and active in your computer
                - Which have Administrator/Superuser privileges
                - User Passwords policy. (empty Passwords, complexity, expiration, etc.)
            - **Firewall configuration**
                - Is it activated?
                - Client/Server ports communications permitted.
                - Applications allowed

- **Hard-drive Encryption**
    - Is activated?
    - Encryption Algorithm used
    - Encryption Key used (attributes such as key length, etc.)
- **Antivirus software**
    - Is it activated?
    - Is it updated regularly? How often?
- **Backup**
    - Is is configured? Periodicity
    - What content is backed up?
- **Password manager**
    - Are you using one to store your different user/passwords? [Note: If you are not using one, do some research and select one to use.]
    - Provide the security properties of your password manager - Encryption algorithm, File storage, Master password requirements, etc.

Deliverable

For this assignment you will prepare a PDF report with screenshots where you review your configurations on the previous operating systems aspects and propose different improvements based on the previous specific operating system hardening guides.

<u>This is the structure that your report must have:</u>

1. Introduction.
   > Description of your computer - Hardware and Operating System details.

2. Operating System configuration aspects
   > For each of the security configuration aspect described below, talk about current and improved states.

2.1 System Users

2.1.1. Current configuration potential weakness. Explain how this weakness can impact your

computer and the information stored.
2.1.2. Configuration improvement details and rationale. Make sure your explanation targets how the existing weakness/es can be reduced or eliminated.
2.1.3. Include screenshot of the initial password policy.

2.2 Firewall Configuration
2.2.1. Current configuration potential weakness. Explain how this weakness can impact your computer and the information stored.
2.2.2. Configuration improvement details and rationale.Make sure your explanation targets how the existing weakness/es can be reduced or eliminated.
2.2.3. Include screenshot of the firewall configuration and firewall status.

2.3 Hard-drive Encryption
2.3.1. Current configuration potential weakness. Explain how this weakness can impact your computer and the information stored.
2.3.2. Configuration improvement details and cryptographic algorithms offered by your system.
2.2.3. Include screenshot of the hard-drive encryption configuration.

2.4 Antivirus software
2.4.1. Current configuration potential weakness.Explain how this weakness can impact your computer and the information stored.
2.4.2. Configuration improvement details and rationale.Make sure your explanation targets how the existing weakness/es can be reduced or eliminated.

2.5 Information back-up
2.5.1. Current configuration potential weakness. Explain how this weakness can impact your computer and the information stored.
2.5.2. Configuration improvement details and rationale. Make sure your explanation targets how the existing weakness/es can be reduced or eliminated.
2.5.3. Include screenshot of the backup tool configuration of your system

2.6 Password manager
2.6.1 Provide the security properties of your password manager: encryption algorithm, storage location, master password requirements, etc. Also provide a rationale behind password managers improving the security posture of the system
-----> If you are not using one, do some research of your options, select one and answer 2.6.1.

3. Conclusion

> Description about the overall security posture that the computer had and how that posture can be improved after your hardening process.

- In the following link you can find a template for the report (link)

Grading Rubric for Assignment 1.1 (1)

Criteria	Pts
This criterion is linked to a learning outcomePart 1 - Introduction Has described their Computer (Hardware and OS Details)	0.5 pts
This criterion is linked to a learning outcomePart 2 - OS Configuration 1. System Users: 1.1. Student has included the number of users, users with superuser or admin privileges and the current password policy of the account. The student has included a photo of the initial password policy. The student has identified the potential weakness of the current configuration and explained how it can impact their computer and the information stored. 1.2. The student has identified and explained how the existing weakness/es can be reduced or eliminated. 1.3. The student has included a screenshot of the initial password policy.	1.5 pts
This criterion is linked to a learning outcomePart 2 - OS Configuration 2. Firewall Configuration: 2.1. The student has identified the current configuration of the firewall and its potential weaknesses. The student has described the importance of a firewall and its use. 2.2.The student has taken a screenshot of the incoming / outgoing ports and connections along with the applications allowed.	1.5 pts
This criterion is linked to a learning outcomePart 2 - OS Configuration 3. Hard Drive Encryption: 3.1.The student has identified and described methods for hard drive encryption and identified the cryptographic algorithms offered by their system, including relevant cryptographic attributes (key length). 3.2.The student has given some information on how encrypting the hard drive could result in reducing the attack surface of their device.	1.5 pts

Criteria	Pts
This criterion is linked to a learning outcomePart 2 - OS Configuration 4. Antivirus Software 4.1.The student has identified and described an antivirus that can be used in their system. 4.2.The student has given some information on if the antivirus they are using is updated regularly or not.	1.5 pts
This criterion is linked to a learning outcomePart 2 - OS Configuration 5. Backup 5.1.The student has mentioned if their data is backed up regularly or not and has identified the potential weakness of not having a backup. 5.2. The student has identified a possible backup system.	1.5 pts
This criterion is linked to a learning outcomePart 2 - OS Configuration 6. Password Manager 6.1.The student has given information about the security properties of their password manager by giving details about its encryption algorithm, file storage and master password requirements. 6.2.The student has mentioned how password managers improve the security posture on their device and all accounts on it.	1.5 pts
This criterion is linked to a learning outcomePart 3 - Conclusion The student should provide a detailed comparison of the changes made in each aspect of the assignment, emphasizing the previous configurations, the new configurations, and the rationale behind the improvements. For each change, they should explain why the adjustment was necessary and how it strengthens the system's security.	0.5 pts

Total points: 10